GREENPULSE MANUFACTURING PLANT #4

Comprehensive Plant Master Reference & Operational Manual

Facility Code: FAC-IND-PUN-004	**Contracted Demand:** 2,500.0 kW
Location: Chakan MIDC Phase II, Pune	**Utility Account:** MSEDCL-HT-9482103
Operating Mode: 3 Shifts / 24x7 Continuous	**Substation Feeder:** 33 kV Express Feeder #3
Effective Date: October 2026	**Compliance:** ISO 50001 / BEE PAT VI / IEEE 519

1. Facility Profile & Electrical Infrastructure

GreenPulse Plant #4 is an advanced precision engineering and heavy component manufacturing facility. The plant operates on a 24x7 three-shift basis with a total connected load of approximately 3,450 kW and a utility-sanctioned Contract Demand of **2,500.0 kW** supplied via a dedicated 33 kV substation transformer.

Electrical Parameters & Power Ingress:

- **Primary Ingress:** 33 kV stepped down to 415V, 3-phase, 50 Hz via dual 2,000 kVA ONAN transformers.
- **Contract Demand (CD):** 2,500.0 kW (Billing demand is maximum of actual kVA or 85% of CD).
- **Tariff Structure:** MSEDCL HT-1 Industrial with Time-of-Day (ToD) tariff. Base tariff: ■7.20/kWh.
- **Peak Hours (18:00–22:00 IST):** 1.44x multiplier (effective ■10.37/kWh). Peak demand charges are strictly tracked.
- **Off-Peak Night Hours (22:00–06:00 IST):** 0.85x incentive multiplier (effective ■6.12/kWh).
- **Power Factor Rebate / Penalty:** Target PF > 0.98. Rebate up to 7% for PF ≥ 0.995; penalty of 2% for every 0.01 drop below 0.90.
- **Backup Generation:** 2 x 1,250 kVA Caterpillar Diesel Generators with automatic transfer switch (ATS).

2. Executive Dashboard KPIs & Live Telemetry Metrics

The GreenPulse AI executive dashboard synthesizes real-time Modbus SCADA telemetry every 15 seconds. The baseline operational benchmarks monitored on the dashboard are structured as follows:

Dashboard Metric	Nominal Value / Range	Threshold / Alert Rule	Operational Impact
Active Facility Demand	1,420 – 1,650 kW	Warning > 2,200 kW Critical > 2,375 kW	Prevents DISCOM 150% penalty on demand exceedance
Daily Energy Draw	18,500 – 22,000 kWh	Baseline variance > ±10%	Tracks production specific energy consumption
System Power Factor	0.96 – 0.99 lagging	Alert if PF < 0.95 Critical if PF < 0.90	Secures utility rebate; prevents reactive surcharge
Voltage THD (IEEE 519)	2.1% – 3.8%	Limit ≤ 5.0% THD_V	Protects electronic drives, motors, and transformer life

Current TDD (IEEE 519)	4.5% – 6.2%	Limit ≤ 8.0% TDD	Minimizes neutral conductor heating and line loss
Fleet Health Index	88.4% Average	Critical if asset < 75%	Identifies bearing and thermal degradation early
Carbon Intensity	0.82 kg CO■e / kWh	National CEA Benchmark	Mandatory BRSR & ESG Scope 2 disclosure

3. Machine Fleet Registry & Digital Twin Specifications

The digital twin tracks 9 primary industrial assets across 6 physical factory zones. Each machine is monitored via IoT edge gateways measuring power (kW), voltage, current, vibration (mm/s RMS), and bearing temperature (°C).

Asset Code	Machine Description & Model	Zone / Bay	Rated / Peak kW	Health / Status
CNC-01	5-Axis Milling Center *DMG Mori DMU 65 monoBLOCK*	Precision Bay 1	22.0 kW / 35 kW	96% / Running
CNC-02	Heavy-Duty 4-Axis Lathe *Doosan Puma 2600SY II*	Precision Bay 2	18.0 kW / 26 kW	94% / Running
COMP-01	Screw Air Compressor #1 (Lead) *Atlas Copco GA 75 VSD+*	Utilities Block A	58.0 kW / 412 kW*	71% / Warning
COMP-02	Variable Speed Compressor #2 *Ingersoll Rand R55i VSD*	Utilities Block A	58.0 kW / 75 kW	92% / Running
FURN-01	Continuous Annealing Furnace *Seco/Warwick Vector 2.0*	Thermal Bay	95.0 kW / 650 kW*	84% / Running
HVAC-01	Central Chiller & Air Handler *Daikin Pathfinder AWV 15*	Chiller Plant	15.0 kW / 180 kW*	88% / Running
PAINT-01	Automated Spray Booth & Plenum *Eisenmann E-Shuttle 200*	Finishing Bay B	32.0 kW / 45 kW	90% / Idle
PUMP-01	Coolant Recirculation Pump *Grundfos CR 64-3 Industrial*	Utilities Block B	12.0 kW / 18 kW	95% / Running
ASSY-01	Robotic Assembly Cell *ABB IRB 6700 Cell*	Assembly Hall	90.0 kW / 115 kW	98% / Running

* Note: Peak instantaneous loads include inductive startup cycles and thermal heating bank elements during full multi-stage operation.

4. Industrial AI & Machine Learning Architecture

GreenPulse AI employs an integrated suite of scikit-learn and XGBoost models running via the in-memory **ModelRegistry** singleton. Models operate with zero per-request disk reloading to ensure sub-100ms inference times.

4.1. 24-Hour Multi-Horizon Energy Load Forecasting (XGBoost):

• **Algorithm:** Multi-stage gradient-boosted decision trees (XGBoost Regressor) trained on multi-season historical plant SCADA data.
• **Features:** Rolling lag power, outdoor ambient temperature, shift schedule phase, production tonnage forecast, and calendar seasonality.
• **Explainability:** TreeExplainer SHAP (SHapley Additive exPlanations) generates real-time feature attribution bar graphs.
• **Accuracy & Metrics:** Mean Absolute Percentage Error (MAPE) = 3.82%, Root Mean Squared Error (RMSE) = 48.6 kW, Confidence = 94.2%.

4.2. Unsupervised Anomaly Detection (Isolation Forest):

• **Algorithm:** High-dimensional Isolation Forest with an empirical contamination factor of 0.03.
• **Telemetry Inputs:** Active power (kW), reactive power (kVAR), 3-phase current imbalance, bearing temperature (°C), and vibration RMS (mm/s).
• **Incident Attribution:** Root causes automatically mapped to ISO 10816 vibration zones, motor phase unbalance, or thermal insulation breakdown.

5. Prescriptive Energy Conservation Measures (ECMs)

The Prescriptive Recommendation Engine ranks facility energy projects using Multi-Criteria Decision Analysis (MCDA). The current prioritized portfolio of active Energy Conservation Measures includes:

ECM Title	Target Asset	Annual Savings	Cost / Payback	Carbon Offset
VFD Retrofit & Pressure Setpoint Optimization (7.5 to 6.8 bar)	COMP-01 Utilities	48,000 kWh/yr **■3,45,600/yr**	■2,35,000 8.2 months	39.4 MT CO■e/yr
Flue Gas Waste Heat Recovery (WHR) System Installation	FURN-01 Thermal Bay	72,000 kWh/yr **■5,18,400/yr**	■6,25,000 14.5 months	59.0 MT CO■e/yr
Chilled Water Supply Temperature Reset (+1.5 °C Reset)	HVAC-01 Chiller Plant	18,500 kWh/yr **■1,33,200/yr**	■15,000 1.4 months	15.2 MT CO■e/yr
Automated Capacitor Bank Stage Tuning (PF 0.96 to 0.995)	Main Substation Plant-Wide	Rebate Capture **■1,80,000/yr**	■45,000 3.0 months	12.0 MT CO■e/yr
Compressed Air Ultrasonic Acoustic Leakage Remediation (22 Leaks)	Utilities Ring Blocks A & B	31,200 kWh/yr **■2,24,640/yr**	■38,000 2.0 months	25.6 MT CO■e/yr

6. What-If Parametric Simulation & Scenario Modeling

The What-If Simulator module empowers plant leadership to model counterfactual operational and economic scenarios before implementing physical setpoint modifications. The core parameters supported include:

• **Production Throughput Scaling:** Simulates facility energy draw when production load varies from 50% to 150% of nominal 320 tons/day capacity.
• **Tariff Optimization & Peak Shift:** Calculates cost variance when thermal annealing or machining shifts are transferred from peak (18:00–22:00) to off-peak night windows.
• **Renewable Energy Integration:** Models the net import load curves and peak shaving impacts of an on-site 850 kWp solar PV array.
• **Equipment Duty Cycle Adjustments:** Simulates the effects of staggering lead/lag compressors (COMP-01 and COMP-02) to prevent simultaneous startup spikes.

7. GHG Protocol Carbon Accounting & ESG Reporting

GreenPulse AI implements automated greenhouse gas emissions accounting aligned with the GHG Protocol Corporate Standard and India's SEBI Business Responsibility and Sustainability Reporting (BRSR) framework.

• **Scope 1 (Direct Emissions):** 12.4 Metric Tons (MT) CO■e / month. Derived from natural gas consumption in the annealing furnace (FURN-01) and intermittent testing of diesel generators.

• **Scope 2 (Indirect Emissions):** 48.6 Metric Tons (MT) CO■e / month. Calculated using the Central Electricity Authority (CEA) grid emission factor of **0.82 kg CO■e / kWh** applied to utility electricity consumption.
• **Total Plant Carbon Footprint:** ~61.0 MT CO■e / month (Annualized ~732 MT CO■e).
• **Specific Energy Consumption (SEC):** 68.5 kWh / Metric Ton of finished industrial output.
• **Net-Zero 2030 Roadmap:** Target of 42% carbon intensity reduction by 2030 through rooftop solar PV expansion (850 kWp), compressed air electrification, and waste heat recovery.

8. Standard Operating Procedures (SOPs) & Regulatory Compliance

Operational procedures are indexed in the RAG vector store for instant retrieval by the AI Copilot. Key protocols and standards governing Plant #4 include:

• **GP-SOP-ENG-04 (Peak Demand Curtailment):**
- *Stage 1 (90% / 2,250 kW):* Automated warning notification to Shift Supervisor. Halt auxiliary testing.
- *Stage 2 (95% / 2,375 kW):* Throttle secondary trim air compressor (COMP-02) and shed comfort air conditioning.
- *Stage 3 (98% / 2,450 kW):* Interlock heating elements on furnace FURN-01 to eliminate immediate penalty risk.

• **GP-SOP-ENG-07 (Power Factor & Harmonics):**
- Daily inspection of APFC microprocessor relay display.
- Verification that THD_V remains < 5.0% and capacitor bank contactor operating temperatures remain < 55 °C.
- Immediate replacement of detuned reactor steps if current harmonics exceed 8% TDD.

• **ISO 10816-3 (Vibration Severity Guidelines):**
- Zone A (< 1.4 mm/s RMS): Newly commissioned state.
- Zone B (1.4 – 2.8 mm/s RMS): Nominal unrestricted long-term operation.
- Zone C (2.8 – 4.5 mm/s RMS): Unsatisfactory operation; plan bearing grease / realignment.
- Zone D (> 4.5 mm/s RMS): Critical vibration; initiate controlled machinery shutdown.

• **ISO 50001:2018 (Energy Management Systems):**
- Energy baseline (EnB) regression models updated quarterly against production tonnage.
- Significant Energy Uses (SEUs): Compressed air supply (28.4%), Furnace thermal heat (38.2%), Precision machining (18.6%).

9. SCADA Telemetry Architecture & Enterprise RBAC Governance

• **SCADA Gateway:** Moxa industrial edge controller streaming Modbus TCP (Port 502) and OPC UA (Port 4840) telemetry.
• **Sync Cycle:** 15-second polling interval with zero-data-loss local circular buffer caching.
• **Role-Based Access Control (RBAC):**
- *Plant Head:* Full executive oversight, setpoint authorizations, financial and compliance sign-offs.
- *Energy Manager:* Complete energy explorer access, tariff modeling, ECM approvals, and carbon disclosures.
- *Maintenance Manager:* Equipment diagnostics, vibration spectrum analysis, work orders, and health indices.
- *Operator:* Restricted bay-level telemetry (assigned machines only); plant-wide financial metrics masked.